

A noncommutative 2-adic Fourier system and an obstruction to the strict doubled-character model

Candidate solution packet for arXiv:2506.14431, Question 5.17

11 August 2026

Status. Candidate full dichotomy, likely valid. The affirmative construction answers the source question under four explicit Fourier-system axioms distilled from the surrounding paper. The source does not itself define the requested term, so the semantic match remains a point for human review.

1 Source question

Jiao, Luo, Zhao, and Zhou ask the following on page 31, Question 5.17, of [1]:

Recall that the construction of noncommutative Walsh and Vilenkin systems appear in the series of the papers by Sukochev and his collaborators. It is natural to ask whether we can construct a noncommutative 2-adic integer system (the classical version is in Example 2.7)?

We conclude this paper with two questions.

Question 5.16. It is an interesting problem to consider Theorem 1.1 for unbounded Vilenkin system. We also point out here that for unbounded Vilenkin system even in the classical setting Theorem 1.1 (i) is a long time open problem. The best related result was proved by Gát [23] for so-called rarely unbounded Vilenkin groups.

Question 5.17. Recall that the construction of noncommutative Walsh and Vilenkin systems appear in the series of the papers by Sukochev and his collaborators [1, 7, 16, 15, 61]. It is natural to ask whether we can construct a noncommutative 2-adic integer system (the classical version is in Example 2.7)?

Acknowledgement. Yong Jiao is partially supported by the NSFC (No. 12125109, No. 11961131003). Sijie Luo is partially supported by the NSFC (No. 12201646), Natural Science Foundation Hunan (No. 2023JJ40696). Dejian Zhou is partially supported by NSFC (No. 12001541), Natural Science Foundation Hunan (No. 2023JJ20058), CSU Innovation-Driven Research Programme (No. 2023CXQD016).

Figure 1: Full-width crop of Question 5.17 on page 31 of [1].

The paper does not formally define “noncommutative 2-adic integer system.” Its noncommutative Vilenkin system has four central structural features: it lives in the hyperfinite II_1 factor, consists of a complete orthonormal family of unitaries, is closed projectively under multiplication, and is built from the classical character data. We formulate the answer using precisely those features. We then prove that the stricter version obtained by doubling the classical character group, exactly as in the Vilenkin construction, is impossible.

2 Definitions

Let

$$K = \mathbb{Z}_2, \quad A = \widehat{K} = \mathbb{Z}[1/2]/\mathbb{Z}.$$

Thus A is the Prüfer 2-group. If $r = a/2^N + \mathbb{Z} \in A$ and $x \in K$, define

$$\chi_r(x) = \exp\left(2\pi i \frac{a(x \bmod 2^N)}{2^N}\right).$$

This is independent of the representative and identifies A with the character group of K .

Definition 1 (Structural 2-adic Fourier system). A structural noncommutative 2-adic Fourier system is a finite factor $(\mathcal{M}, \tau)$ together with a countable family of unitaries $\{X_{r,n} : r \in A, n \in \mathbb{Z}\}$ such that

- (i) the family is an orthonormal basis of $L_2(\mathcal{M}, \tau)$;
- (ii) products of basis elements are scalar multiples of basis elements;
- (iii) $\{X_{r,0} : r \in A\}$ is the classical character system of $\mathbb{Z}_2$;
- (iv) $\mathcal{M}$ is the hyperfinite II_1 factor.

The integer index records powers of one noncommuting translation. The strict doubled-character version would instead ask for a projective unitary basis indexed by $A \times A$, mirroring the doubled dual group in the source's noncommutative Vilenkin construction.

3 Proof intuition

Translation by 1 is the odometer on $\mathbb{Z}_2$. Its action on a character is diagonal: translating χ_s multiplies it by the scalar $e^{-2\pi i s}$. The crossed product therefore turns the ordinary characters into explicit noncommuting Fourier monomials $\chi_r u^n$. Orthogonality is read directly from the canonical crossed-product trace.

The only point not immediate from the multiplication formula is that the ambient algebra really is the hyperfinite II_1 factor. Dyadic residue classes form exact odometer towers. Matrix units obtained by moving between levels of a tower give nested copies of M_{2^N} . Their cyclic shift differs from the true odometer unitary only on the top level, whose trace is 2^{-N} ; hence the difference tends to zero in L_2 .

The strict obstruction is dual to this mechanism. Two copies of the Prüfer group are too divisible to support a nontrivial scalar commutator pairing: their dual is $\mathbb{Z}_2^2$, which contains no nonzero divisible subgroup. The integer translation direction in the positive construction is exactly what escapes that obstruction.

4 Affirmative construction

Theorem 2 (Odometer 2-adic system). *Let $\alpha(x) = x + 1$ on $K = \mathbb{Z}_2$, and form the group-measure-space von Neumann algebra*

$$\mathcal{M} = L_\infty(K) \rtimes_\alpha \mathbb{Z}.$$

Let u be the canonical implementing unitary, with $ufu^ = f \circ \alpha^{-1}$. For $(r, n) \in A \times \mathbb{Z}$, set*

$$X_{r,n} = \chi_r u^n.$$

Then $\mathcal{M}$ is the hyperfinite II_1 factor and $\{X_{r,n}\}_{(r,n) \in A \times \mathbb{Z}}$ is an orthonormal basis of $L_2(\mathcal{M})$. Moreover,

$$X_{r,n} X_{s,m} = e^{-2\pi i n s} X_{r+s, n+m}. \tag{1}$$

The subfamily $\{X_{r,0} : r \in A\}$ is precisely the classical character system of the 2-adic integers, and the whole system is noncommutative.

Proof. The character identity $\chi_s(x-1) = e^{-2\pi is}\chi_s(x)$ gives

$$u^n \chi_s u^{-n} = e^{-2\pi in s} \chi_s.$$

Equation (1) follows. Taking $n = 1$ and $s = 1/2 + \mathbb{Z}$ gives $u\chi_s = -\chi_s u$, so the system is genuinely noncommutative.

The canonical trace on the algebraic crossed product is

$$\tau\left(\sum_{k \in \mathbb{Z}} f_k u^k\right) = \int_K f_0 d\mu.$$

Consequently,

$$\langle X_{r,n}, X_{s,m} \rangle = \tau(X_{r,n}^* X_{s,m}) = \delta_{n,m} \int_K \overline{\chi_r} \chi_s d\mu = \delta_{n,m} \delta_{r,s}.$$

The algebraic crossed product is L_2 -dense in $\mathcal{M}$, while the characters $\{\chi_r : r \in A\}$ are an orthonormal basis of $L_2(K)$. Approximating each coefficient f_k by characters proves completeness of the displayed family.

We next verify the factor and hyperfinite assertions. Translation by 1 is free: if $x+n = x$ in $\mathbb{Z}_2$ for an integer n , then $n = 0$. It is ergodic as well. Indeed, if $f \circ \alpha = f$, then its Fourier coefficient at r obeys

$$\widehat{f}(r) = e^{2\pi ir} \widehat{f}(r),$$

and $e^{2\pi ir} \neq 1$ for every nonzero $r \in A$. Hence f is constant. The usual Fourier-coefficient computation in the crossed product is also short here: if a central element has coefficients f_n , commutation with $L_\infty(K)$ and freeness force $f_n = 0$ for $n \neq 0$; commutation with u then makes f_0 α -invariant. Thus the center is scalar and $\mathcal{M}$ is a II_1 factor.

For a direct proof of hyperfiniteness, fix $N \geq 1$, put $d = 2^N$, and let

$$C_j^{(N)} = j + d\mathbb{Z}_2, \quad p_j^{(N)} = \mathbf{1}_{C_j^{(N)}} \text{quad}(0 \leq j < d).$$

Define

$$e_{ij}^{(N)} = p_i^{(N)} u^{i-j} p_j^{(N)}.$$

Since $u^k p_j^{(N)} u^{-k} = p_{j+k}^{(N)}$ with indices modulo d , these elements obey

$$e_{ij}^{(N)} e_{kl}^{(N)} = \delta_{j,k} e_{il}^{(N)}, \quad (e_{ij}^{(N)})^* = e_{ji}^{(N)}.$$

They therefore span a matrix algebra $F_N \cong M_d$. Splitting each residue class modulo d into its two residue classes modulo $2d$ gives

$$e_{ij}^{(N)} = e_{ij}^{(N+1)} + e_{i+d,j+d}^{(N+1)}, \tag{2}$$

so $F_N \subset F_{N+1}$.

The diagonal of $\bigcup_N F_N$ contains every dyadic cylinder function and is therefore L_2 -dense in $L_\infty(K)$. Finally define the cyclic shift

$$v_N = \sum_{j=0}^{d-2} e_{j+1,j}^{(N)} + e_{0,d-1}^{(N)} \in F_N.$$

The unitaries u and v_N agree on $1 - p_{d-1}^{(N)}$. Hence

$$\|u - v_N\|_2 = \|(u - v_N)p_{d-1}^{(N)}\|_2 \leq 2\tau(p_{d-1}^{(N)})^{1/2} = \frac{2}{\sqrt{d}} \rightarrow 0.$$

Thus the L_2 -closure of $\bigcup_N F_N$ contains $L_\infty(K)$ and u , so it is all of $L_2(\mathcal{M})$. Therefore $\mathcal{M}$ is hyperfinite. Being a separable hyperfinite II_1 factor, it is $\mathcal{R}$. All four defining properties are now proved. $\square$

5 The strict doubled-character model is impossible

Theorem 3 (Doubled-Prüfer obstruction). *Let $G = A \times A$. Suppose unitaries $\{W_g : g \in G\}$ in a von Neumann algebra satisfy*

$$W_g W_h = \sigma(g, h) W_{g+h} \quad (g, h \in G)$$

for scalars $\sigma(g, h) \in \mathbb{T}$. Then all the W_g commute. In particular, such a family cannot be an orthonormal basis generating a noncommutative factor.

Proof. The scalar commutator

$$c(g, h) = \frac{\sigma(g, h)}{\sigma(h, g)}$$

satisfies $W_g W_h = c(g, h) W_h W_g$. Associativity, or a direct commutator calculation, shows that c is a bicharacter. Thus

$$\Phi : G \longrightarrow \widehat{G}, \quad \Phi(g)(h) = c(g, h),$$

is a group homomorphism.

The group $A = \mathbb{Z}[1/2]/\mathbb{Z}$ is divisible, hence so is G . Its Pontryagin dual is $\widehat{G} = \mathbb{Z}_2 \times \mathbb{Z}_2$. The image of a divisible group under a homomorphism is divisible. But $\mathbb{Z}_2^2$ has no nonzero divisible subgroup: if y belongs to such a subgroup, then for every k it is divisible there by 2^k , so

$$y \in \bigcap_{k \geq 0} 2^k \mathbb{Z}_2^2 = \{0\}.$$

Therefore $\Phi = 0$, so $c(g, h) = 1$ for all g, h . The unitaries commute. If their linear span were weakly dense, the generated von Neumann algebra would be abelian, not a factor. $\square$

Corollary 4 (No nested carry-Weyl bases). *There is no sequence of projective unitary bases of M_{2^N} indexed by $(\mathbb{Z}/2^N \mathbb{Z})^2$ which is compatible, up to phases, with unital matrix-algebra inclusions and with the classical character inclusions $a \mapsto 2a$ in both coordinates.*

Proof. A compatible sequence would yield in the inductive limit a projective unitary family indexed by

$$\varinjlim_N (\mathbb{Z}/2^N \mathbb{Z})^2 = A^2.$$

Each finite projective matrix basis has a nondegenerate commutator pairing: an element in its radical would give a nonidentity central basis unitary in a full matrix algebra. The limit would therefore contain noncommuting unitaries, contradicting Theorem 3.

Equivalently, at the finite level every injection into $(\mathbb{Z}/2^{N+1} \mathbb{Z})^2$ has image inside (indeed equal to) twice that group. Pulling an alternating pairing back through two doubled coordinates multiplies its coefficient by four; relative to denominator 2^{N+1} this becomes an even, hence degenerate, coefficient modulo 2^N . $\square$

6 Interpretation and limitations

Theorem 2 gives an explicit affirmative construction with the basic Fourier-system features used around the source question. It also explains why the natural index group is $A \times \mathbb{Z}$: the pairing $(n, s) \mapsto e^{-2\pi i n s}$ between integer odometer translations and 2-adic characters is nondegenerate. Replacing the integer direction by a second divisible copy of A kills every possible scalar commutator by Theorem 3.

The result does *not* assert that some one-dimensional enumeration of $\{X_{r,n}\}$ has the same Cesàro maximal inequalities as the Vilenkin ordering in [1]. Nor can the basis be ordered so that its canonical finite initial blocks are nested cyclic Weyl bases: Corollary 4 is exactly the obstruction. Those are analytic or ordering requirements not stated in Question 5.17.

7 Verification and novelty status

The proof uses no computational or unproved dependency. The multiplication, trace orthogonality, factor property, and matrix approximation have been checked independently line by line. The strict no-go proof reduces to the elementary fact that $\mathbb{Z}_2^2$ has no nonzero divisible subgroup.

A bounded literature search on 11 August 2026 used the exact question phrase, the source title and arXiv id, the author names, and combinations of “noncommutative 2-adic integer system,” “odometer crossed product,” “Prüfer group twisted group algebra,” and “noncommutative solenoid.” No paper explicitly answering Question 5.17 was found. Noncommutative p -solenoids in [2, 3] twist $\mathbb{Z}[1/p]^2$, not the doubled classical character group $(\mathbb{Z}[1/p]/\mathbb{Z})^2$, so they are adjacent rather than exact answers. The odometer and crossed-product ingredients are standard; the identification with Question 5.17 and the doubled-Prüfer obstruction were not found in the bounded search. Novelty confidence is therefore moderate.

Human-review recommendation. Verify (i) that the structural definition captures the authors’ intended undefined term, (ii) the convention in the phase of (1), (iii) the nesting identity (2) and L_2 approximation of u , and (iv) the claim that every scalar commutator of an A^2 -projective family lands in $\widehat{A^2} = \mathbb{Z}_2^2$. If these checks pass, retain this as a full affirmative construction plus a full no-go theorem for the strict model.

References

- [1] Y. Jiao, S. Luo, T. Zhao, and D. Zhou, *Almost uniform convergence for noncommutative Vilenkin-Fourier series*, arXiv:2506.14431, 2025.
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- [3] F. Latrémolière and J. Packer, *Noncommutative solenoids and their projective modules*, arXiv:1311.1193, 2013.